

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: <u>Book</u>		
Property Name	Data Type	Source Tag
<u>bookId</u>	<u>int</u>	<u>System Generated</u>
<u>title</u>	<u>string</u>	<u>User Input</u>
<u>author</u>	<u>string</u>	<u>User Input</u>
<u>genre</u>	<u>string</u>	<u>User Input</u>
<u>totalCopies</u>	<u>int</u>	<u>User Input</u>
<u>availableCopies</u>	<u>int</u>	<u>Default Value</u>
<u>borrowCount</u>	<u>int</u>	<u>Default Value</u>

Entity Name: <u>Member</u>		
Property Name	Data Type	Source Tag
<u>memberId</u>	<u>int</u>	<u>System Generated</u>
<u>memberName</u>	<u>string</u>	<u>User Input</u>
<u>phoneNumber</u>	<u>string</u>	<u>User Input</u>
<u>email</u>	<u>string</u>	<u>User Input</u>
<u>registrationDate</u>	<u>string</u>	<u>System Generated</u>
<u>activeLoansCount</u>	<u>int</u>	<u>Calculated</u>
<u>isActive</u>	<u>bool</u>	<u>Default Value</u>

Entity Name: <u>Loan</u>		
Property Name	Data Type	Source Tag
<u>loanId</u>	<u>int</u>	<u>System Generated</u>
<u>memberId</u>	<u>int</u>	<u>From List</u>
<u>bookId</u>	<u>int</u>	<u>From List</u>
<u>borrowDate</u>	<u>string</u>	<u>System Generated</u>
<u>dueDate</u>	<u>string</u>	<u>Calculated</u>
<u>returnDate</u>	<u>string</u>	<u>System Generated</u>
<u>fineAmount</u>	<u>decimal</u>	<u>Calculated</u>

Entity Name: <u>Reservation</u>		
Property Name	Data Type	Source Tag
<u>reservationId</u>	<u>int</u>	<u>System Generated</u>
<u>memberId</u>	<u>int</u>	<u>From List</u>
<u>bookId</u>	<u>int</u>	<u>From List</u>
<u>reservationDate</u>	<u>string</u>	<u>System Generated</u>
<u>notificationSent</u>	<u>bool</u>	<u>Default Value</u>
<u>status</u>	<u>string</u>	<u>Default Value</u>
<u>queuePosition</u>	<u>int</u>	<u>Calculated</u>

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

A public library system allows people to register as members and borrow books. Each book has information such as title, author, genre, and number of available copies. When a member borrows a book, the system creates a loan record and gives the member 14 days to return it. If the member returns the book late, the system automatically calculates a fine based on the overdue days. If all copies of a book are already borrowed, members can reserve the book and will be notified when it becomes available again. The librarian can also use the system to view active loans, check overdue books, and generate reports showing the most borrowed books.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	RegisterMember	Creates a new library member
2	AddBook	Adds a new book to the library
3	BorrowBook	Creates a loan for a member
4	ReturnBook	Processes a returned book
5	CalculateFine	Calculates overdue fines
6	ReserveBook	Creates a reservation for unavailable books
7	NotifyReservedMember	Searches for books using title keywords
8	ViewActiveLoans	Displays all active loans
9	ViewOverdueLoans	Shows overdue loans sorted by delay
10	GenerateMostBorrowedBooks Report	Shows the most borrowed books

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
guestId	int	<u>System Generated</u>
guestName	string	<u>User Input</u>
guestPhone	string	<u>User Input</u>
guestEmail	string	<u>User Input</u>
guestNationality	string	<u>User Input</u>
registrationDate	string	<u>System Generated</u>

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	<u>System Generated</u>
roomNumber	string	<u>User Input</u>
roomType	string	<u>User Input</u>
floorNumber	int	<u>User Input</u>
nightlyPrice	decimal	<u>User Input</u>
isAvailable	bool	<u>Default Value</u>

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	<u>System Generated</u>
guestId	int	<u>From List</u>
roomId	int	<u>From List</u>
checkInDate	string	<u>User Input</u>
checkOutDate	string	<u>User Input</u>
numberOfNights	int	<u>Calculated</u>
totalPrice	decimal	<u>Calculated</u>
status	string	<u>Default Value</u>

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	System Generated
bookingId	int	From List
guestId	int	Calculated
roomId	int	Calculated
actualCheckIn	string	System Generated
actualCheckOut	string	System Generated
finalAmount	decimal	Calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

A guest first registers in the hotel system and gets a unique ID. The hotel stores rooms with their type, price, and availability. When a guest books a room, the system creates a booking record connecting the guest and the room. The guest selects the check-in and check-out dates, while the system automatically calculates the number of nights and total price. After booking, the room becomes unavailable. When the guest arrives, the system creates a check-in record and saves the actual check-in time. When the guest checks out, the system records the checkout time, calculates the final amount, and makes the room available again. If the booking is cancelled before check-in, the booking status changes to cancelled and the room becomes available immediately.

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	RegisterGuest	Creates a new guest account
2	AddRoom	Adds a room to the hotel system
3	ViewAvailableRooms	Displays all available rooms
4	CreateBooking	Creates a room booking
5	CalculateTotalPrice	Calculates booking total cost
6	CancelBooking	Cancels a booking
7	CheckInGuest	Creates a check-in record

#	Service Name	What it does
8	CheckOutGuest	Processes guest checkout
9	UpdateRoomAvailability	Changes room availability status